

- It supports C/C++ based APIs for extended Python for Numpy's data structures and computations.
- All the following mathematical operations are supported – random number generation, Fourier transforms, linear algebra and matrix operations.

To import and start using the NumPy library, use the following command:

```
import numpy as np
```

Pandas

Pandas is one of the most popular Python libraries for data manipulation and data cleaning. Often, Pandas is used with NumPy and SciPy along with analytical libraries to build data visualisation. Pandas has the following built-in capabilities:

- It supports the matplotlib library.
- It supports Scikit-learn.
- It supports array based functions and computing.
- In addition to NumPy, Pandas is used for homogeneous numerical array data types.

Pandas Series: A Series is a one-dimensional array. This is almost similar to an object containing a sequence of values (only similar data types are allowed) along with the index having its labels.

In the following example, there are four integer data types in the array 'my_data' and this data type is shown below as 'int64'.

```
1 my_data = pd.Series([1,9,-5,18])
1 my_data

0    1
1    9
2   -5
3   18
dtype: int64
```

Various data operations can be performed on Series objects. Just as we can have our own index labels, math

operations like *greater than*, *less than*, *equal to*, etc, can be done.

```
1 my_data2 = pd.Series([1,9,-5,18],
index = ['q', 'w', 'e', 'r'])
my_data2

q    1
w    9
e   -5
r   18
my_data2['w']
9
my_data2['r'] = 98
my_data2
q    1
w    9
e   -5
r   98
dtype: int64

2 my_data2[my_data2 < 100]
q    1
w    9
e   -5
r   98
dtype: int64

1 my_data2 = [my_data2 **2]
1 my_data2
[q    1
 w   81
 e   25
 r  324]
dtype: int64
```

Pandas DataFrames: A DataFrame in Pandas contains an ordered collection of columns. Each can be of a different value type (Boolean, numeric, string or integer). Unlike a Series object, DataFrame will have both row and column indexes. The data will be stored in the one- or two-dimensional array format.

To import and start using the Pandas library, use the following set of

commands and steps:

```
Import pandas as pd
print(my_list)
['shashi', 'dhar', 'soppin', 'is',
'writing', 'this', 'article']
df = pd.DataFrame(my_list)
df

0    shashi
1    dhar
2    soppin
3     is
4   writing
5    this
6   article
```

SciPy

This is a fundamental library for scientific computing. This library provides many numerical routines that are efficient. There are some routines for numerical integration, optimisation, linear algebra and statistical purposes.

SciKit-learn

SciKit-learn is one of the most used ML libraries in Python. It provides simple and efficient tools for carrying out data analysis with data mining related features. The following are the major groups of algorithms that are supported in the SciKit-learn library:

- Classification
- Regression
- Dimensionality reduction
- Model selection
- Preprocessing
- Clustering 

References

- [1] <https://numpy.org/>
- [2] <https://pandas.pydata.org/>
- [3] <https://www.scipy.org/>
- [4] <https://scikit-learn.org/stable/>
- [5] <https://matplotlib.org/>

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